



## **Lab Report-04 (Option-1)**

**Course:** CISC5950

**Title:** Big Data Computing

### ***Submitted To:***

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**Date of Submission:** 05/11/2026

## P1: Toxic Comment Classification

### Dataset

The Jigsaw Toxic Comment Classification dataset contains approximately 560,000 Wikipedia comments, each labeled across six toxicity categories: toxic, severe\_toxic, obscene, threat, insult, and identity\_hate. We created a binary label (toxic if any category is flagged) for classification.

### How the Code Works

The code reads training data from HDFS, preprocesses text by lowercasing and removing non-alphabetic characters, then builds a five-stage Spark ML Pipeline. The Tokenizer splits comment text into individual words. StopWordsRemover filters out common English words that don't carry meaning. HashingTF converts the filtered word list into a sparse term-frequency vector with 10,000 features using the hashing trick. IDF (Inverse Document Frequency) rescales the term frequencies to weight rare, more informative words higher. Finally, Logistic Regression classifies the weighted feature vector as toxic or non-toxic.

The model is trained on an 80/20 split of the training data for validation, then evaluated on the actual test set (test.csv joined with test\_labels.csv, filtering out unlabeled rows marked with -1).

### Results

| Metric             | Value  |
|--------------------|--------|
| AUC-ROC            | 0.8906 |
| Accuracy           | 0.9188 |
| Weighted Precision | 0.9080 |
| Weighted Recall    | 0.9188 |
| F1 Score           | 0.9030 |

```
MODEL EVALUATION RESULTS
AUC-ROC:          0.8906
Accuracy:         0.9188
Weighted Precision: 0.9080
Weighted Recall:  0.9188
F1 Score:         0.9030
```

*Fig. 1: P1 Evaluation Results*

## P2: Heart Disease Prediction

### Dataset

The Framingham Heart Study dataset contains approximately 4,240 patient records with 16 columns including demographic data (age, sex), behavioral factors (currentSmoker, cigsPerDay), medical history (BPMeds, prevalentStroke, prevalentHyp, diabetes), clinical measurements (totChol, sysBP, diaBP, BMI, heartRate, glucose), and the target variable TenYearCHD indicating 10-year coronary heart disease risk.

### How the Code Works

This task converts the sample Python code provided in the lab specification to Spark ML. The code reads framingham.csv as strings (to handle 'NA' values), drops the education column as per the sample code, replaces 'NA' strings with null, then drops rows with missing values (about 12% of the dataset). All columns are cast to double, and VectorAssembler combines the 14 remaining features into a single vector. Logistic Regression is trained with maxIter=100 and regParam=0.01. The model reports the top risk factors by coefficient magnitude and evaluates on a held-out test set.

### Results

| Metric             | Value  |
|--------------------|--------|
| AUC-ROC            | 0.7373 |
| Accuracy           | 0.8360 |
| Weighted Precision | 0.7995 |
| Weighted Recall    | 0.8360 |
| F1 Score           | 0.7739 |

The model is heavily biased toward predicting no disease due to class imbalance (approximately 85% negative cases). The confusion matrix shows 575 true negatives, 4 false positives, 110 false negatives, and 6 true positives.

| MODEL EVALUATION RESULTS |        |
|--------------------------|--------|
| AUC-ROC:                 | 0.7373 |
| Accuracy:                | 0.8360 |
| Weighted Precision:      | 0.7995 |
| Weighted Recall:         | 0.8360 |
| F1 Score:                | 0.7739 |

*Fig. 2: P2 Evaluation Results*

### **P3: Census Income Classification (Logistic Regression)**

#### **Dataset**

The UCI Adult Census Income dataset contains 32,561 training records and 16,281 test records with 14 attributes including age, workclass, education, marital status, occupation, relationship, race, sex, capital gain/loss, hours per week, and native country. The target variable is binary: income >50K or <=50K. The data contains a mix of categorical (8 columns) and numerical (6 columns) features, plus missing values marked with '?'.

#### **How the Code Works**

The code handles several data quality challenges. The test file has a junk first line ('1x3 Cross validator') which is skipped by reading via RDD, filtering by index, then converting back to DataFrame. Income labels in the test file end with a period (<=50K.) which is stripped using `regexp_replace`. Rows with '?' values are filtered out.

For feature engineering, 8 categorical columns (workclass, education, marital\_status, occupation, relationship, race, sex, native\_country) are processed through `StringIndexer` to convert string categories to numeric indices, then `OneHotEncoder` to create binary feature vectors. These encoded vectors are combined with 6 numeric features (age, fnlwgt, education\_num, capital\_gain, capital\_loss, hours\_per\_week) using `VectorAssembler`. The complete pipeline feeds into `Logistic Regression` with `maxIter=100`.

#### **Results**

| Metric             | Value  |
|--------------------|--------|
| AUC-ROC            | 0.9012 |
| Accuracy           | 0.8461 |
| Weighted Precision | 0.8392 |

|                 |        |
|-----------------|--------|
| Weighted Recall | 0.8461 |
| F1 Score        | 0.8398 |

The confusion matrix shows 10,585 true negatives, 775 false positives, 1,542 false negatives, and 2,158 true positives.

```
LOGISTIC REGRESSION - EVALUATION RESULTS
AUC-ROC:          0.9012
Accuracy:         0.8461
Weighted Precision: 0.8392
Weighted Recall:   0.8461
F1 Score:         0.8398
```

*Fig. 3: P3 Evaluation Results*

# P4: Census Income (Random Forest & Decision Tree)

## How the Code Works

P4 reuses the same data loading and feature engineering pipeline from P3 (imported via shared functions). It trains two additional classifiers on the same Census Income data: Random Forest Classifier with 100 trees and max depth 10, and Decision Tree Classifier with max depth 10. Both use the same StringIndexer, OneHotEncoder, and VectorAssembler pipeline. The code also reports feature importance rankings from the Random Forest model.

## Model Comparison

| Metric    | Logistic Reg (P3) | Random Forest | Decision Tree |
|-----------|-------------------|---------------|---------------|
| AUC-ROC   | 0.9012            | 0.9041        | 0.7119        |
| Accuracy  | 0.8461            | 0.8489        | 0.8446        |
| Precision | 0.8392            | 0.8420        | 0.8371        |
| Recall    | 0.8461            | 0.8489        | 0.8446        |
| F1 Score  | 0.8398            | 0.8400        | 0.8368        |

## Analysis

Random Forest achieves the best performance across all metrics, with AUC-ROC of 0.9041 compared to Logistic Regression's 0.9012 and Decision Tree's 0.7119. While all three models achieve similar accuracy (84-85%), the significant gap in AUC-ROC for Decision Tree (0.71) indicates it produces poorly calibrated probability estimates compared to the other two methods.

Random Forest's advantage comes from ensemble averaging: by combining predictions from 100 independently trained decision trees, it reduces overfitting and variance compared to a single Decision Tree. Logistic Regression performs nearly as well as Random Forest on this dataset because the decision boundary between income classes is largely linear with respect to the encoded features.

```
DECISION TREE - EVALUATION RESULTS
AUC-ROC:          0.7119
Accuracy:         0.8446
Weighted Precision: 0.8371
Weighted Recall:  0.8446
F1 Score:         0.8368
```

*Fig. 4: P4 Decision Tree Evaluation Results*

```
RANDOM FOREST - EVALUATION RESULTS
AUC-ROC:          0.9041
Accuracy:         0.8489
Weighted Precision: 0.8420
Weighted Recall:  0.8489
F1 Score:         0.8400
```

*Fig. 5: P4 Random Forest Evaluation Results*

| MODEL COMPARISON SUMMARY |               |               |
|--------------------------|---------------|---------------|
| Metric                   | Random Forest | Decision Tree |
| AUC-ROC                  | 0.9041        | 0.7119        |
| Accuracy                 | 0.8489        | 0.8446        |
| Weighted Precision       | 0.8420        | 0.8371        |
| Weighted Recall          | 0.8489        | 0.8446        |
| F1 Score                 | 0.8400        | 0.8368        |

*Fig. 6: P4 Model Comparison Summary*

### **Video Demonstration**

<https://drive.google.com/file/d/1gjxxDaEwAbxWZ-0Br5iwO9B1RKG7i5kh/view?usp=sharing>